1.	$\int\left(\frac{1}{2}x^3+\frac{3}{\sqrt{x}}-4\right)\mathrm{d}x=\frac{1}{2}\times\frac{x^4}{4}+3\times\frac{x^{\frac{1}{2}}}{\frac{1}{2}}-4x+c$ $=\frac{1}{8}x^4+6x^{\frac{1}{2}}-4x+c$	M1, A1 A1, A1 **(4 marks)**

2. (a)	$5(x+3) > 4(2x-5) \Rightarrow 5x+15 > 8x-20 \Rightarrow ax > b$ or $px < q$	M1
	$\Rightarrow x < \frac{35}{3}$	A1
		(2)
(b) (i)	$x^2 - 6x + 1 = (x-3)^2 \pm ... = (x-3)^2 - 8$	M1, A1
(ii)	$(x-3)^2 - 8 = 0 \Rightarrow x = 3+\sqrt{8}$ or $3-\sqrt{8}$	M1
	$x^2 - 6x + 1 \geqslant 0 \Rightarrow x \leqslant 3-\sqrt{8}, x \geqslant 3+\sqrt{8}$	A1
		(4)
(c)	$x \leqslant 3-\sqrt{8},\ 3+\sqrt{8} \leqslant x < \frac{35}{3}$	B1
		(1)
		(7 marks)

3.(a)	$\left(\frac{dy}{dx}\right) = 3x^2 + \frac{48}{\sqrt{x}}$	M1, A1, A1 **(3)**
(b)	$\left(\frac{d^2y}{dx^2}\right) = 6x - \frac{24}{x^{\frac{3}{2}}}$ $\Rightarrow x^{\frac{5}{2}} = \ldots \qquad x = 2^{\frac{4}{5}}$	M1 dM1 A1 **(3)** **(6 marks)**

4.(a)	Sets $kx+2=\frac{2}{x}+3x-4$ and attempts to collect terms or multiply through by x	M1
	$(k-3)x+6-\frac{2}{x}=0 \Rightarrow (k-3)x^2+6x-2=0$ *	A1*
		(2)
(b)	Attempts b^2-4ac for $(k-3)x^2+6x-2=0$	M1
	Solves $b^2-4ac=0$ for $(k-3)x^2+6x-2=0 \Rightarrow 36+8(k-3)=0 \Rightarrow k=...$	dM1
	$k=-\frac{3}{2}$	A1
		(3)
		(5 marks)

5 (a)	$9x^3-10x^2+x=x\left(9x^2-10x+1\right)=x(ax\pm1)(bx\pm1)$ with $ab=9$ $=x(9x-1)(x-1)$	M1 A1 **(2)**
(b)	States or implies that $x=3^y$ AND sets = to their 1 or $\frac{1}{9}$ Solves their $3^y=\text{"1"}$ or $3^y=\text{"}\frac{1}{9}\text{"}$ $y=0,-2$	M1 dM1 A1 **(3)** **(5 marks)**

6 (a)	Angle $AFB = \dfrac{\pi - 2.275}{2} = 0.433$ *	B1*
		(1)
(b)	Attempts $r\theta = 6.2 \times 2.275 = (14.105)$	M1
	Attempts $\left(x^2\right) = 6.4^2 + 6.2^2 - 2 \times 6.4 \times 6.2\cos(0.433) = (7.36)$ or $(x = 2.71)$	M1
	Correct attempt = $2x + r\theta + 12.8 = 2 \times 2.714 + 14.105 + 12.8 = 32.3\,(\text{m})$	dM1, A1
		(4)
(c)	Attempts $\dfrac{1}{2}r^2\theta = \dfrac{1}{2} \times 6.2^2 \times 2.275 = (43.7255)$	M1
	Attempts $\dfrac{1}{2}ab\sin C = \dfrac{1}{2} \times 6.2 \times 6.4 \times \sin 0.433 = (8.325)$	M1
	Correct attempt = $2 \times 8.325 + 43.7255 = 60.4\left(\text{m}^2\right)$	dM1, A1
		(4)
		(9 marks)

7 (a)	States or implies that B is $(10, 6)$	B1
	$\left(AB^2\right) = (10-0)^2 + ("6"-2)^2 \Rightarrow AB = 2\sqrt{29}$	M1, A1
		(3)
(b)	States or implies that grad $AB = \frac{2}{5}$	B1
	Uses perpendicular gradient rule $\Rightarrow$ grad $l_2 = -\frac{5}{2}$	M1
	$y - "6" = "-\frac{5}{2}"(x-10) \Rightarrow 5x + 2y - 62 = 0$	dM1, A1
		(4)
(c)	C is $\left(\frac{62}{5}, 0\right)$	B1 ft
	$\left(BC^2\right) = \left(\frac{62}{5} - 10\right)^2 + ("6"-0)^2 \Rightarrow BC = \frac{6\sqrt{29}}{5} = (6.46)$	
	Area $ABCD = 69.6$	M1, A1
		(3)
		(10 marks)

8 (a)

$f'(x) = 2(x-3)(3x+2) = 6x^2 - 14x - 12$ B1

$f(x) = 2x^3 - 7x^2 - 12x + k$ M1, A1

Uses $P(4, 13) \Rightarrow 13 = 2\times64 - 7\times16 - 12\times4 + k \Rightarrow k = ...$ M1

$(f(x)) = 2x^3 - 7x^2 - 12x + 45$ A1

(5)

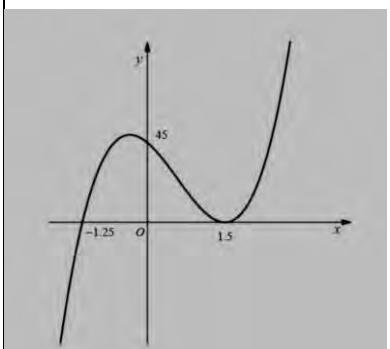
(b)

$2x^3 - 7x^2 - 12x + 45 \equiv (x^2 - 6x + 9)(px + q) \Rightarrow p = "2"$ or $q = \frac{"45"}{9}$ B1 ft

States either $2x^3 - 7x^2 - 12x + 45 \equiv (x-3)^2(2x+5)$ or $p = 2$ & $q = 5$ following a fully correct (a) or correct but incomplete part (a) B1

(2)

(c)



+ ve cubic curve B1

"Correct" y intercept for their part (b) equation. So $9q$ or "45" B1ft

Turning point at (1.5, 0) B1

x intercept at $(-1.25, 0)$ B1

(4)

(11 marks)

9. (a)	(i) 3 (ii) 101 (iii) 2	B1 B1 B1 **(3)**
(b)	$a = \frac{2\pi}{3}$ o.e.	B1 **(1)**
(c)	Attempts x coordinate of Q: $x = "\frac{2\pi}{3}" + \pi = \frac{5\pi}{3}$	M1
	Attempts both y coordinates $P\left(\frac{2\pi}{3}, \frac{\pi}{3}\right)$ $Q\left(\frac{5\pi}{3}, -\frac{2\pi}{3}\right)$	dM1
	Attempts mid-point $PQ = \left(\frac{1}{2}\left(\frac{2\pi}{3} + \frac{5\pi}{3}\right), \frac{1}{2}\left(\frac{\pi}{3} - \frac{2\pi}{3}\right)\right)$	ddM1
	$= \left(\frac{7\pi}{6}, -\frac{\pi}{6}\right)$	A1 **(4)** **(8 marks)**

10	$y = \frac{2}{3}x^3 - 25x - \frac{56}{x} + \frac{194}{3}$	
(a)	$\frac{dy}{dx} = 2x^2 - 25 + \frac{56}{x^2}$	M1, A1
	Finds $\left.\frac{dy}{dx}\right\|_{x=2} = 2\times 4 - 25 + \frac{56}{4} = -3$	dM1
	Equation of tangent $y + 8 = "-3"(x-2) \Rightarrow y = -3x - 2$ *	ddM1, A1*
		(5)
(b)	Sets $2x^2 - 25 + \frac{56}{x^2} = -3$	M1
	$x^4 - 11x^2 + 28 = 0$	dM1, A1
	$(x^2 - 4)(x^2 - 7) = 0 \Rightarrow x^2 = ..$	ddM1
	$x = \sqrt{7}$ only	A1
		(5)
		(10 marks)